

Structural Admissibility: An I2OS Framework for Secure Generative Systems

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Abstract

We introduce Structural Admissibility, a transition-based safety framework for generative systems. Unlike conventional approaches that rely on post-generation filtering, the proposed I2OS framework constrains execution by permitting only structurally admissible state transitions. This establishes a proactive safety paradigm. The framework integrates admissibility constraints, reality confirmation, and collapse detection. We demonstrate its applicability to LLM hallucination suppression by reformulating hallucination as an unsupported state transition.

1 Introduction

Modern AI systems rely on optimization, prediction, and post-hoc filtering for safety. However, these approaches allow unsafe intermediate states and require reactive correction.

Conventional pipeline:

Generate $\rightarrow$ *Filter*

In contrast, we propose a proactive paradigm:

Execute only if admissible

This work introduces a proactive safety paradigm by constraining execution at the transition level rather than filtering outputs post hoc.

Contributions:

- A transition-based formulation of safety
- A unified admissibility framework integrating reality confirmation and collapse detection
- A structural interpretation of LLM hallucination

2 Related Work

AI safety has been addressed through post-generation filtering, reinforcement learning from human feedback, and constraint-based decoding. Hallucination mitigation methods include retrieval-augmented generation and verification models. These approaches are useful, but they often operate after or during generation rather than defining execution eligibility as a transition-level constraint.

Our approach differs by enforcing admissibility at the transition level, preventing invalid outputs from becoming executable states.

3 Structural Admissibility

A transition is admissible if:

$$C(S_t, T, S_{t+1}) = 1$$

Execution is defined as:

$$GO = \mathbf{1}[C(S_t, T, S_{t+1}) = 1]$$

Principle:

$$Security = NonCollapseConstraint$$

4 I2OS Security Model

$$GO(t) = \mathbf{1}[\Gamma(Q(t'))R(t') > \theta \wedge Reality(t) = 1 \wedge \neg Collapse(t)]$$

4.1 Components

- Q : Question generation represented as a semantic gradient
- R : Structural resonance
- $Reality$: External validation of the transition
- $Collapse$: Structural instability detection

5 Architecture

6 Experimental Evaluation

6.1 Setup

State representation:

$$S = (F, \Sigma, E, \phi, P)$$

Baseline:

$$GO_{baseline} = \mathbf{1}[Pr > \theta]$$

Proposed:

$$GO_{proposed} = \mathbf{1}[C \wedge Reality \wedge \neg Collapse]$$

6.2 Results

Table 1: Baseline vs Proposed Execution

Method	Entries	Validity	Failure	Uncertainty
Baseline	High	Low	High	Poor
Proposed	Low	High	Low	Explicit

Structural Admissibility Architecture

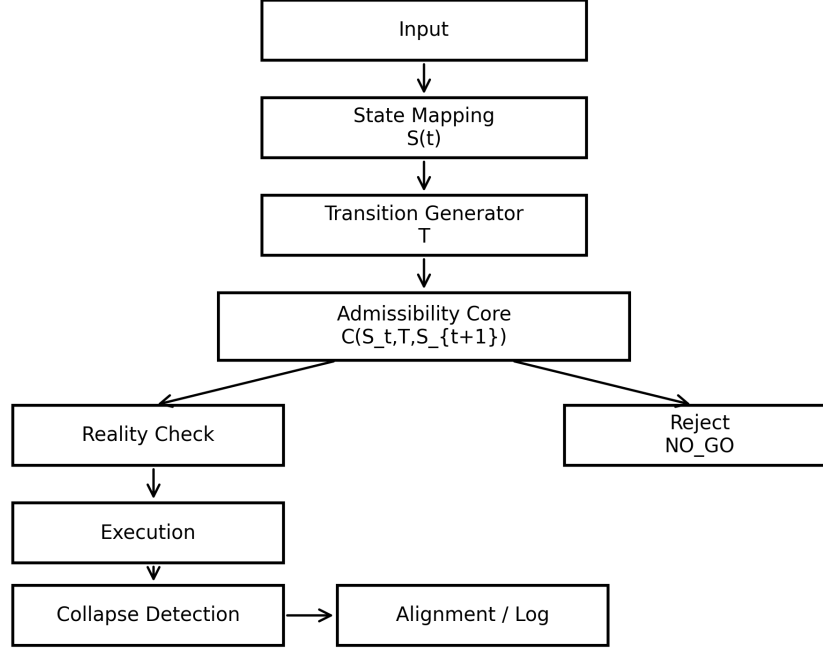


Figure 1: Structural Admissibility Architecture

6.3 Observation

Execution is not equivalent to structural alignment:

$$Execution = Admissible\ Transition$$

7 LLM Hallucination Suppression

Hallucination is reformulated as:

$$Hallucination = Unsupported\ Transition$$

Admissibility constraints:

$$C_{LLM} = \mathbf{1}[C_{context} \wedge C_{source} \wedge C_{causality} \wedge C_{scope} \wedge C_{uncertainty}]$$

Execution modes:

- Admissible $\rightarrow$ Full response
- Partial $\rightarrow$ Qualified response
- Invalid $\rightarrow$ Refusal

Conventional vs I2OS Safety

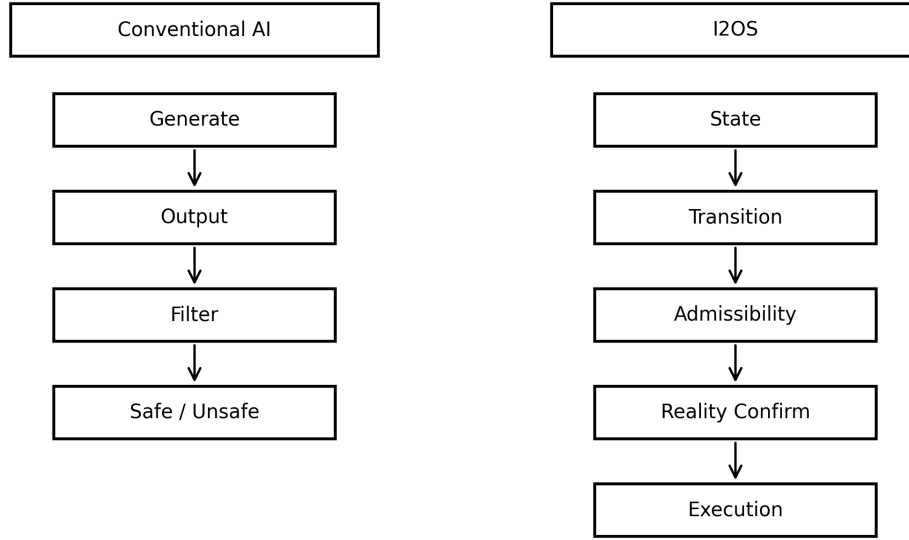


Figure 2: Conventional vs I2OS Safety

8 Discussion

This framework shifts safety from output filtering to transition-level validation. It reframes AI safety as the prevention of invalid state transitions rather than the correction of outputs after generation.

9 Limitations

- Requires explicit constraint definition
- Domain dependency of admissibility criteria
- Additional computational overhead
- Current evaluation is conceptual and requires larger empirical validation

10 Conclusion

Structural Admissibility establishes a proactive safety paradigm:

$$Security = AdmissibleTransitionOnly$$

Admissibility Funnel

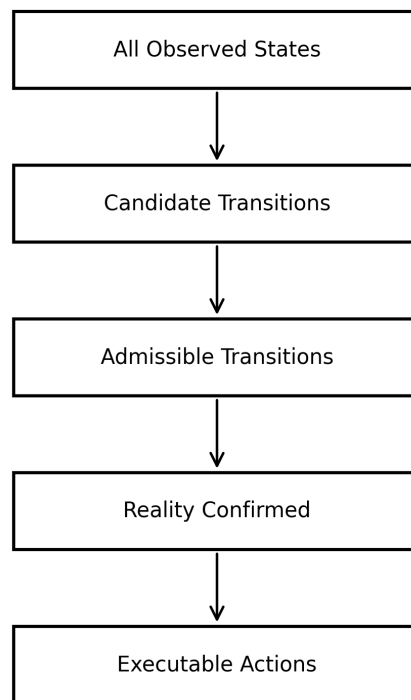


Figure 3: Admissibility Funnel

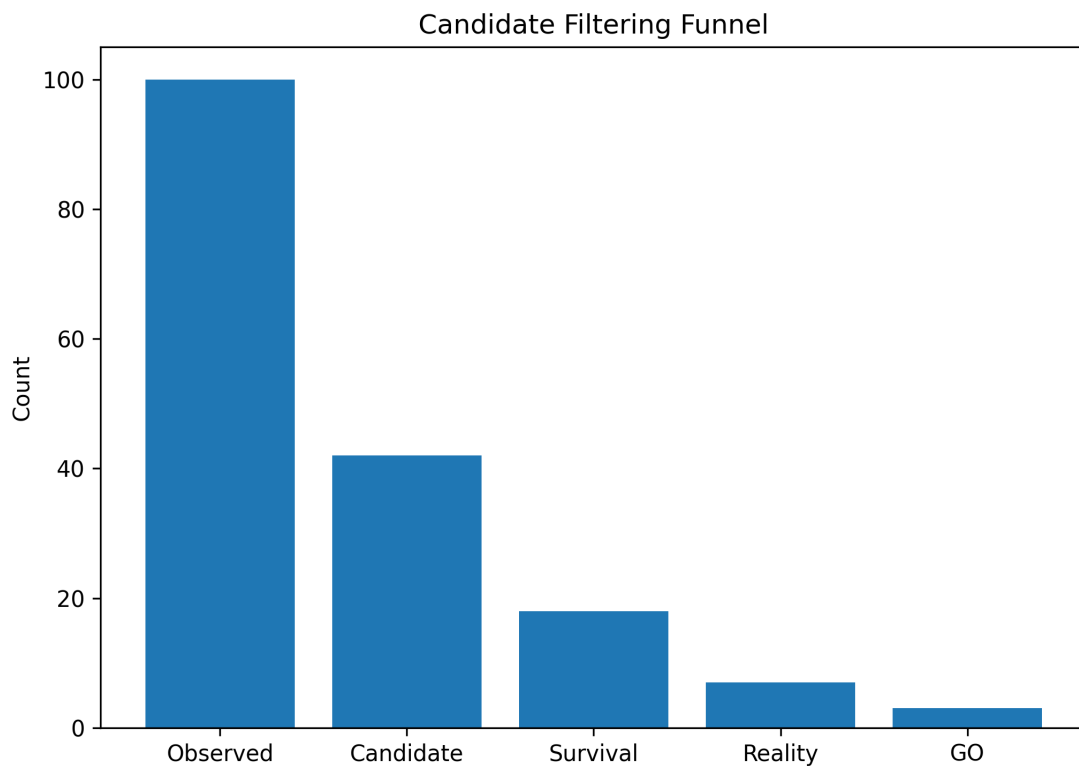


Figure 4: Candidate Filtering Funnel

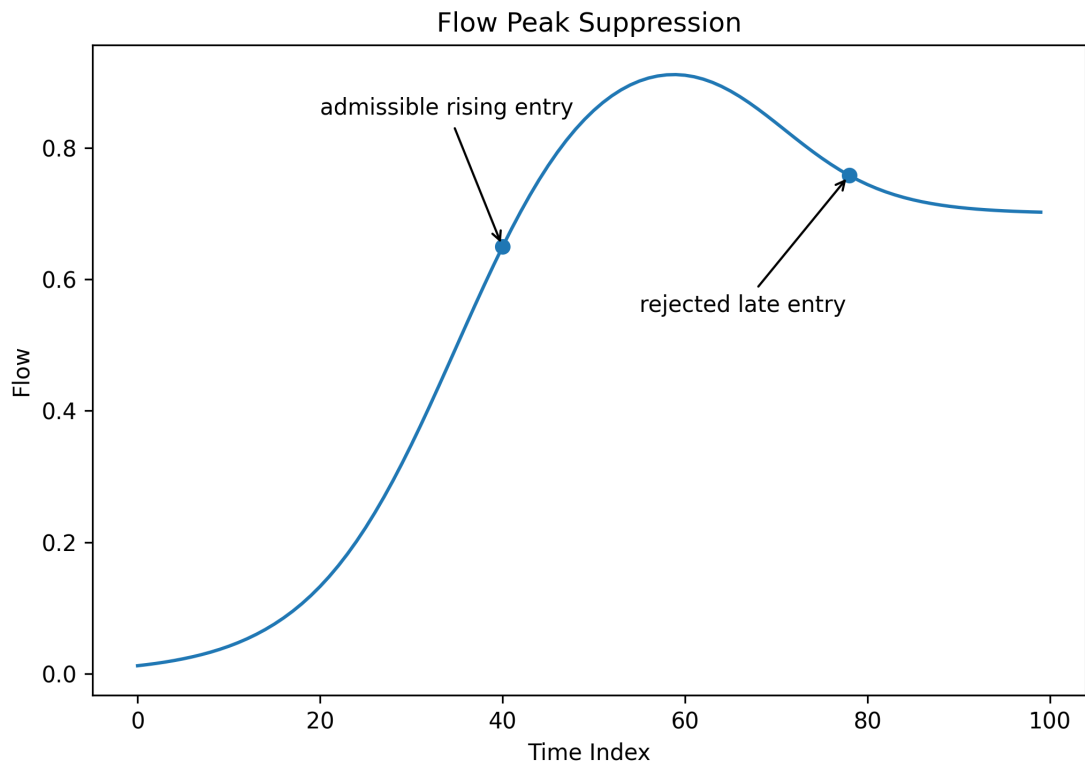


Figure 5: Flow Peak Suppression

LLM Structural Admissibility Pipeline

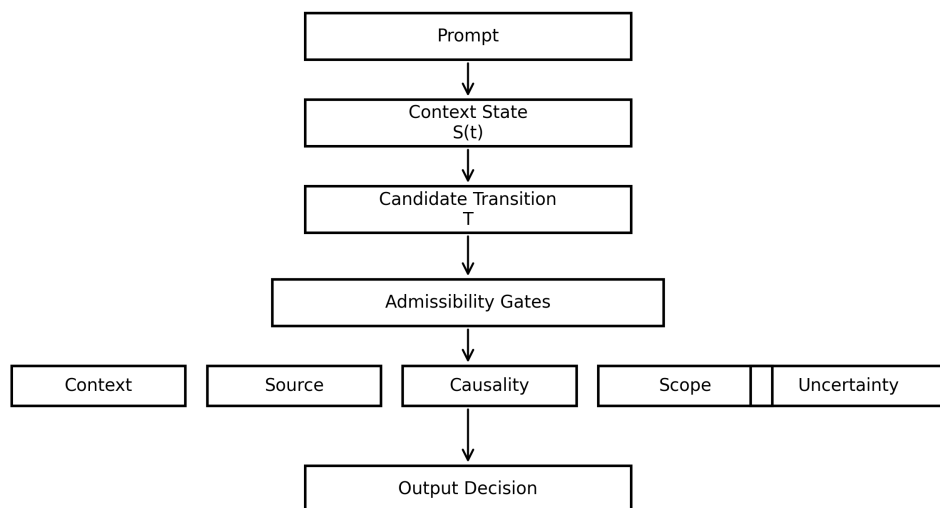


Figure 6: LLM Structural Admissibility Pipeline